

$$\text{Then } (T - \lambda I)\vec{y} = (T - \lambda I)^{p_x}(\vec{x}) = \vec{0} \\ \Rightarrow \vec{y} \in E_\lambda.$$

Example Let $A \in \mathbb{C}^{6 \times 6}$ and $P \in \mathbb{C}^{6 \times 6}$ such that $A = PJP^{-1}$ where $J = \left(\begin{array}{ccc|ccc} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{array} \right)_{6 \times 6}$

$$\text{Then } f(x) = \det(A - \lambda I) = \det(J - \lambda I) = (2 - \lambda)^4 (3 - \lambda)^2.$$

$$\text{Eigenvalues: } \lambda = 2 \text{ (multiplicity } = 4), \quad \lambda = 3 \text{ (multiplicity } = 2).$$

$$\text{Denote } P = [\vec{p}_1 \ \vec{p}_2 \ \vec{p}_3 \ \vec{p}_4 \ \vec{p}_5 \ \vec{p}_6]. \text{ Then } A = PJP^{-1} \Leftrightarrow AP = PJ$$

$$\Leftrightarrow A\vec{p}_1 = 2\vec{p}_1 \Leftrightarrow (A - 2I)\vec{p}_1 = \vec{0} \Rightarrow \vec{p}_1 \in \text{Null}(A - 2I)$$

$$A\vec{p}_2 = \vec{p}_1 + 2\vec{p}_2 \Leftrightarrow (A - 2I)\vec{p}_2 = \vec{p}_1 \Rightarrow \vec{p}_2 \in \text{Null}(A - 2I)^2$$

$$A\vec{p}_3 = \vec{p}_2 + 2\vec{p}_3 \Leftrightarrow (A - 2I)\vec{p}_3 = \vec{p}_2 \Rightarrow \vec{p}_3 \in \text{Null}(A - 2I)^3$$

$$A\vec{p}_4 = 2\vec{p}_4 \Leftrightarrow (A - 2I)\vec{p}_4 = \vec{0} \Rightarrow \vec{p}_4 \in \text{Null}(A - 2I)$$

$$A\vec{p}_5 = 3\vec{p}_5 \Leftrightarrow (A - 3I)\vec{p}_5 = \vec{0} \Rightarrow \vec{p}_5 \in \text{Null}(A - 3I)$$

$$A\vec{p}_6 = \vec{p}_5 + 3\vec{p}_6 \Leftrightarrow (A - 3I)\vec{p}_6 = \vec{p}_5 \Rightarrow \vec{p}_6 \in \text{Null}(A - 3I)^2$$

Remarks: 1) $\{\vec{p}_1, \vec{p}_2, \dots, \vec{p}_6\}$ is a set of generalized eigenvectors

$$2). \{\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4\} \in K_2. \quad \dim K_2 = 4 = \text{multiplicity of } \lambda = 2?$$

$$. \{\vec{p}_5, \vec{p}_6\} \in K_2, \quad \dim K_2 = 2 = \text{multiplicity of } \lambda = 3?$$

$$3). \{\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4\} \in \text{Null}(A - 2I)^3 \quad K_2 = \text{Null}(A - 2I)^4?$$

$$\{\vec{p}_5, \vec{p}_6\} \in \text{Null}(A - 3I)^2 \quad K_4 = \text{Null}(A - 3I)^2?$$

$$4). \# \text{ of Jordan blocks with diagonal } \lambda = 2 = 2 = \dim E_2$$

$$\# \text{ of Jordan blocks with diagonal } \lambda = 3 = 1 = \dim E_3. \quad // \text{ end of March 5}$$

Conjectures: Let $T: V \rightarrow V$ be a linear transformation.

1). There exists a basis of V which consists of generalized eigenvectors of T .

2). $K_\lambda = \{ \vec{x} \in V : (T - \lambda I)^{m_\lambda}(\vec{x}) = \vec{0}, \text{ where } m_\lambda \text{ is the multiplicity of the eigenvalue } \lambda \}$

3). $\dim K_\lambda = m_\lambda$, where m_λ is the multiplicity of the eigenvalue λ .

4). The number of Jordan blocks corresponding to $\lambda = \dim E_\lambda$

5). $K_\lambda \cap K_\mu = \{ \vec{0} \}$ when $\mu \neq \lambda$.

Outline of the proof of the main Theorem.

Let $T: V \rightarrow V$ be a linear transformation. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be all the distinct eigenvalues of T . Let $m_1, m_2, \dots, m_k$ be the corresponding multiplicities of $\lambda_1, \dots, \lambda_k$, respectively.

Let $K_{\lambda_1}, K_{\lambda_2}, \dots, K_{\lambda_k}$ be the generalized eigenspaces. Let $E_{\lambda_1}, E_{\lambda_2}, \dots, E_{\lambda_k}$ be the corresponding eigenspaces.

Part 1): Prove that if $B_{\lambda_1}, B_{\lambda_2}, \dots, B_{\lambda_k}$ are basis for $K_{\lambda_1}, K_{\lambda_2}, \dots, K_{\lambda_k}$. Then $B_{\lambda_1} \cup B_{\lambda_2} \cup \dots \cup B_{\lambda_k}$ is a basis for V .

Part 2): Construction of those B_{λ_i} 's in part 1 using 'cycles'.

Definition: Let $T: V \rightarrow V$ be a linear transformation. Let $W \subseteq V$ be a subspace. Then W is called T -invariant if for any $\vec{x} \in W$, $T(\vec{x}) \in W$.

Lemma: For any $a, b \in F$: $(T - aI)(T - bI) = (T - bI)(T - aI)$.

Proposition: Let $T: V \rightarrow V$ be a linear transformation. Let $W \subseteq V$ be a T -invariant subspace. Let

$f_W(x)$ be the characteristic polynomial of $T|_W: W \rightarrow W$. Let $f(x)$ be the characteristic polynomial of $T: V \rightarrow V$. Then $f_W(x) \mid f(x)$, i.e. $f_W(x)$ is a factor of $f(x)$.

Recall: To compute $f_W(x)$, one fixes a basis B_W of W , then compute $[T|_W]_{B_W} = [T]_{B_W}$

If $\dim W = m$, then $[T|_W]_{B_W} \in \mathbb{C}^{m \times m}$. Then $f_W(x) = \det([T|_W]_{B_W} - \lambda I)$

To compute $f(x)$, one fixes a basis B of V , then compute $[T]_B$. If $\dim V = n$ then $[T]_B \in \mathbb{C}^{n \times n}$. Then $f(x) = \det([T]_B - \lambda I)$.

Proof: Suppose $\dim W = m$. Let $B_W = \{\vec{w}_1, \dots, \vec{w}_m\}$ be a basis of W . Let $A = [T|_W]_{B_W} \in \mathbb{C}^{m \times m}$

Denote $A = (a_{ij})$. Then $T\vec{w}_j = \sum_{i=1}^m a_{ij} \vec{w}_i \Rightarrow f_W(x) = \det(A - \lambda I)$

Extend B_W to a basis of V : $B = \{\vec{w}_1, \dots, \vec{w}_m, \vec{w}_{m+1}, \dots, \vec{w}_n\}$.

$$\text{Then } T\vec{w}_j = \sum_{i=1}^m a_{ij} \vec{w}_i, \quad j=1, \dots, m \quad T\vec{w}_k = \sum_{i=1}^n a_{ik} \vec{w}_i$$

$$\Rightarrow [T]_B = \left(\begin{array}{c|c} A & M \\ \hline 0 & C \end{array} \right)$$

$$\Rightarrow f(x) = \det([T]_B - \lambda I) = \det(A - \lambda I) \det(C - \lambda I) = f_W(x) \det(C - \lambda I)$$

$$\Rightarrow f_W(x) \mid f(x).$$

Proposition: Let $T: V \rightarrow V$ be a linear transformation. Let λ be an eigenvalue of T with algebraic multiplicity λ .

1). K_λ is T -invariant

2). K_λ is $(T - \lambda I)$ -invariant

3). K_λ is $(T - \mu I)$ -invariant, $\forall \mu \in \mathbb{C}$

4). If $\mu \neq \lambda$, then by 4). we can define: $(T - \mu I)|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$. Then $(T - \mu I)|_{K_\lambda}$ is one-to-one.

5). If $\mu \neq \lambda$ is another eigenvalue of T , then $K_\lambda \cap K_\mu = \{\vec{0}\}$.

6). $\dim(K_\lambda) \leq m_\lambda$

7). $K_\lambda = \text{Nul}(T - \lambda)^{m_\lambda}$.

Proof: $\forall \vec{x} \in K_\lambda$. Then $\exists p \in \mathbb{Z}^+$ s.t. $(T - \lambda I)^p(\vec{x}) = \vec{0}$.

$$\text{Then 1) } (T - \lambda I)^p(T(\vec{x})) = T((T - \lambda I)^p(\vec{x})) = T(\vec{0}) = \vec{0}$$

$$\Rightarrow T(\vec{x}) \in K_\lambda. \quad \Rightarrow K_\lambda \text{ is } T\text{-invariant.}$$

$$2) (T - \lambda I)^p((T - \lambda I)(\vec{x})) = (T - \lambda I)^{p+1}(\vec{x}) = \vec{0}$$

$$\Rightarrow (T - \lambda I)(\vec{x}) \in K_\lambda \quad \Rightarrow K_\lambda \text{ is } (T - \lambda I)\text{-invariant.}$$

$$3). (T-\lambda I)^p (T-\mu I)\vec{x} = (T-\mu I) (T-\lambda I)^p (\vec{x}) = \vec{0}$$

$$\Rightarrow (T-\mu I)(\vec{x}) \in K_\lambda \Rightarrow K_\lambda \text{ is } (T-\mu I)\text{-invariant.}$$

4). As K_λ is $(T-\mu I)$ -invariant, we consider a restriction map: $T-\mu I|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$.

Suppose, towards a contradiction, $\text{Null}((T-\mu I)|_{K_\lambda}) \neq \{\vec{0}\} \Rightarrow \text{let } \vec{x} \in \text{Null}((T-\mu I)|_{K_\lambda}) \text{ such that } \vec{x} \neq \vec{0}$.

$$(T-\mu I)(\vec{x}) = \vec{0}. \text{ As } \vec{x} \in K_\lambda \exists \text{ smallest } p \in \mathbb{Z}^+, \text{ s.t. } (T-\lambda I)^p(\vec{x}) = \vec{0}.$$

$$\text{By the choice of } p: \vec{y} = (T-\lambda I)^{p-1}(\vec{x}) \neq \vec{0}.$$

$$(T-\lambda I)(\vec{y}) = (T-\lambda I)(T-\lambda I)^{p-1}(\vec{x}) = \vec{0} \Rightarrow \vec{y} \in E_\lambda.$$

$$(T-\mu I)(\vec{y}) = (T-\mu I)(T-\lambda I)^{p-1}(\vec{x}) = (T-\lambda I)^{p-1}(T-\mu I)(\vec{x}) = \vec{0} \in \vec{y} \in E_\mu.$$

$$\text{As } \lambda \neq \mu: E_\lambda \cap E_\mu = \{\vec{0}\}. \Rightarrow \vec{y} = \vec{0} \text{ contradiction.}$$

5). Suppose $\vec{x} \in K_\lambda \cap K_\mu$. $\exists p, q \in \mathbb{Z}^+$ such that $(T-\lambda I)^p(\vec{x}) = \vec{0}$ and $(T-\mu I)^q(\vec{x}) = \vec{0}$.

As $(T-\mu I)|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$ is one-to-one, $(T-\mu I)^q|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$ is also one-to-one

$$\text{As } \vec{x} \in K_\lambda, (T-\mu I)^q(\vec{x}) = \vec{0} \Rightarrow \vec{x} = \vec{0}.$$

6). As m_λ is the multiplicity of λ , which is an eigenvalue of $T: V \rightarrow V$.

Then the characteristic polynomial of T has the form: $f(x) = (\lambda - x)^{m_\lambda} g(x)$ where

$$(\lambda - x) \nmid g(x). \text{ i.e. } g(x) \text{ doesn't have the factor } (\lambda - x).$$

Assume $\dim(K_\lambda) = m$. Since K_λ is T -invariant, we may consider the restriction map:

$$T|_{K_\lambda}: K_\lambda \rightarrow K_\lambda. \text{ let } f_{K_\lambda}(x) \text{ be the characteristic polynomial of } T|_{K_\lambda}.$$

$$\text{then } f_{K_\lambda}(x) \mid f(x).$$

Claim: λ is the only eigenvalue of $T|_{K_\lambda}$ on K_λ . end of Month 7

Proof of the claim: ① λ is an eigenvalue of $T|_{K_\lambda}$ on K_λ . In fact $E_\lambda \subseteq K_\lambda$

$$\Rightarrow \exists \vec{x} \neq \vec{0} \text{ and } \vec{x} \in E_\lambda \subseteq K_\lambda \text{ such that } (T-\lambda I)\vec{x} = \vec{0} \Leftrightarrow (T|_{K_\lambda} - \lambda I)\vec{x} = \vec{0}$$

$$\Rightarrow \vec{x} \text{ is an eigenvector of } T|_{K_\lambda} \text{ on } K_\lambda \text{ associated with eigenvalue } \lambda.$$